

AJAY CHOPRA | E-Mail [ajayrchopra@gmail.com](mailto:ajayrchopra@gmail.com) | Mob.9953477583 | Experience: 6+ years

## OBJECTIVE

Looking for an opportunity to give my best to the work which I am assigned in field of VLSI design and help the organisation to reach their Goals .

## EDUCATION

### Vlsi expert

Course on logic synthesis and STA

DELHI TECHNOLOGICAL UNIVERSITY, Delhi  
Masters in VLSI and Embedded systems

08/2009 to 05/2011  
CGPA 8.9

Guru Gobind Singh Indraprastha University  
Amity school of engineering and technology  
Bachelors of electronics and communication Engineering

08/2005 to 08/2009.  
78 %

## TECHNICAL SKILLS

- Adept with various ASIC flow file formats like **LEF, DEF, GDS, OA, SPEF, DSPF, SPICE**
- Working knowledge of programming languages like **C, C++, SQL, PERL and Assembly language, Verilog** .
- Microprocessor (8085/8086) and Microcontrollers (AT Mega8/8051) .

## WORK EXPERIENCE

Agasthya app labs

project duration

May 2022 to present.

### 1) Worked for client (Innatera ) in Netherland on Physical design domain .

|                 |         |
|-----------------|---------|
| Technology      | 28nm    |
| No of instances | 70k     |
| Frequency       | 800Mhz  |
| No of macros    | 70k     |
| Tool            | INNOVUS |

2) Client is Broadcom working as lead Physical design engineer. Role here is to do floorplan, placement, CTS, Routing and physical verification.

3) Blocks worked on

4) Pcore\_bmt\_d\syn\_top

|                 |          |
|-----------------|----------|
| Technology      | 5nm      |
| No of instances | 1million |
| Frequency       | 1.1 Ghz  |
| No of macros    | 30       |
| Tool            | INNOVUS  |

### 5) D2d\_axi\_ctrl\_bridge

|                 |          |
|-----------------|----------|
| Technology      | 5nm      |
| No of instances | 5million |
| Frequency       | 1.2 Ghz  |
| No of macros    | 60       |
| Tool            | INNOVUS  |

### 6) hbm3\_mc4\_syn\_wrap

|                 |          |
|-----------------|----------|
| Technology      | 5nm      |
| No of instances | 2million |
| Frequency       | 80Mhz    |
| No of macros    | 40       |
| Tool            | INNOVUS  |

### 7) scu\_top\_syn\_wrap

|                 |                |
|-----------------|----------------|
| Technology      | 5nm            |
| No of instances | 8million       |
| Frequency       | 850Mhz         |
| No of macros    | 70             |
| Tool            | <b>INNOVUS</b> |

**Mirafra Technologies**  
**Client intel**

**Project duration**

**May 2021-May 2022**

**Block name :** DHM design                      14nm design  
Cell count 150  
Did one technique of low power designing (multi voltage domain) .

**Block name** Par\_pcie4                      5nm design  
Par Pcie4 controller was there along with the cpu, memories and switches

|                 |                 |
|-----------------|-----------------|
| Technology      | 5nm             |
| No of instances | 10million       |
| Frequency       | 1.2 Ghz         |
| No of macros    | 80              |
| Tool            | Fusion Compiler |

Cell count 10 million  
Did floorplan, placement, CTS, routing and Physical verification. Also did manual routing to meet the resistance and timing goals. Check legality and create placement. Also removed shorts and opens. Written scripts to automate the DRC violations.

**Member of technical staff, Digicomm Technologies**    project duration

**Dec 2019 to May 2021**

Block : CABO HM                      7nm design

- Client Qualcomm.
- My role was to run complete PD flow (NBF) and check if there is any issue while running the flow and check if it's a design issue or tool issue.
- Technology 7nm.
- Block used : CABO HM.
- Tool used : **INNOVUS**.

**ADVANCED ENGINEER, ALTRAN TECHNOLOGIES**    project duration

**OCT 2018 to OCT 2019**

**Block name :** ADDMULT and MINI CPU TOP                      7nm design

- Client INTEL.
- Technology 7nm and 16 nm .
- Worked on inway flow (c16) and cheetah flow(N7).
- Block used :Blocks which I worked on were one was **Addmult** and second was **Mini CPU top** (hierarchical block ).
- TOOLS USED: Synopsys icc2,QRC extraction tool.
- Cheetah flow is a GUI based environment and scripts were written in TCL with variable mentioned in GUI .
- Checked complete **RTL to GDS flow along with physical verification (FLOOR PLANNING, PLACEMENT, CTS, Routing, STA, Physical verification(DRC LVS) )**.
- TOOL ICC2

**Associate Manager HCL Technologies**

**project duration**

**April 2016-OCT2018**

**SOFTWARE PROFILE**

- As part of software defined Networking team worked on Brocade controller and open day light controller and switches from Brocade, Pica 8 3297 .work include deployment and configuration of switches.
- Along with testing of SDN worked on automation of Networking jobs like ping, SSH, telnet, reporting the results into an xls format using scripting.

**Lecturer (Delhi Technological University)**

project duration

**August 2015 to April 2016**

#### **Lecturer**

- I have taught to B.Tech and M.Tech students different subjects like Digital System Design, Verilog, VHDL, Fault tolerance including different algorithm's like PODEM, D-Algorithms, BIST, DFT and PIC and ARM microcontrollers.

**Product validation Engineer II Cadence Design Systems, Noida project duration June 2011 to Dec 2014**

Worked on different designs from ST,TI technology 28nm, 45nm and 90nm designs .

- Developing test strategy for features of parasitic extraction tool **QRC**.
- Technology 28 nm,45 nm,90 nm
- Benchmarking the Accuracy of QRC tool.
- Developing test cases to effectively validate the new features in QRC.
- Handling key customers like Texas Instrumentation, STMicroelectronics, Freescale, Fujitsu and Qualcomm.
- Review of functional specification.
- Tool Used **Encounter**

#### **ACADEMIC PROJECTS**

2009-PRESENT

- OTRA and its Application (M. Tech project) Designing of QTRA and checking its functionality i.e characteristics, specification are being met or not .
- Digital Clock that has Normal, Seconds, Alarm & Sleep Display (B. Tech project) .
- Temperature Display using LM 35 Sensor Which Gives Temperature in Degree Celsius (B. Tech project).
- Automatic Traffic Light Control System (B. Tech project).
- Parallel Port Interfacing (using Turbo C) (B. Tech project).
- Designing of CMOS Two Stage Compensated OP-AMP Using PSPICE (VLSI Project) (M. Tech project).
- Designing of "Comparator" using PSPICE (VLSI) (M. Tech project).
- Designing of Triple port RAM using Verilog (DKOP project) .